Aharonov–Bohm Effect: The Quantum Mechanical Perspective

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Abstract. This study note establishes the foundational mathematical and physical machinery of gauge invariance in classical electrodynamics and quantum mechanics, serving as a critical prerequisite for investigating non-local quantum phenomena such as the Aharonov-Bohm effect. We first review the classical framework of electromagnetic fields, analyzing how the vector and scalar potentials undergo local gauge transformations while preserving the invariance of physical observables. This configuration provides the precise analytic criteria required to couple electromagnetic potentials to quantum mechanical wavefunctions via minimal coupling. We contrast the structural failure of the Time-Dependent Schrödinger Equation under isolated potential transformations with its exact form invariance when paired with a compensating local phase shift of the wavefunction. Finally, we outline how these local phase factors resolve mathematical gauge discrepancies and physically dictate the modulation of quantum interference patterns within field-free regions.

Contents

1. Classical Electrodynamics

Recall that in electrodynamics, the force experienced by a charged particle is described by:

$$F = q(\mathbf{v} \times \mathbf{B}) \tag{1}$$

We introduce potentials as a method to obtain the fields E and B. Recall:

$$\mathbf{E} = \nabla\phi - \frac{\partial A}{\partial t} \tag{2}$$

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{3}$$

2. Electromagnetic Interactions in Quantum Mechanics

The Hamiltonian for a charged particle q and momentum $\mathbf{p}$ in the presence of electromagnetic fields is:

$$H = \frac{1}{2m}(\mathbf{p} - q\mathbf{A})^2 + q\varphi, \tag{4}$$

whereϕ and $\mathbf{A}$ are identified in (2) and (3) respectively. Substituting $\mathbf{p} \longrightarrow -i\hbar\nabla$ into (4) yields the operator:

$$\hat{H} = \frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2 + q\varphi, \tag{5}$$

Which can be expanded to yield:

$$\begin{aligned}
\hat{H} &= \frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2 + q\varphi \\
&= \frac{1}{2m}(-i\hbar\nabla - q\mathbf{A}) \cdot (-i\hbar\nabla - q\mathbf{A}) + q\varphi \\
&= \frac{1}{2m}\left[-\hbar^2\nabla^2 + i\hbar q\nabla \cdot \mathbf{A} + i\hbar q\mathbf{A} \cdot \nabla + q^2\mathbf{A}^2\right] + q\varphi \\
&= \frac{1}{2m}\left[-\hbar^2\nabla^2 + i\hbar q(\nabla \cdot \mathbf{A}) + 2i\hbar q\mathbf{A} \cdot \nabla + q^2\mathbf{A}^2\right] + q\varphi \\
&= -\frac{\hbar^2}{2m}\nabla^2 + \frac{i\hbar q}{2m}(\nabla \cdot \mathbf{A}) + \frac{i\hbar q}{m}\mathbf{A} \cdot \nabla + \frac{q^2}{2m}\mathbf{A}^2 + q\varphi
\end{aligned}$$

3. Gauge Invariance and Transformations for Scalar and Vector Potentials in Classical Electrodynamics

In quantum mechanics, we introduce the idea of gauge transformations. However, the underlying reason behind *gauge*–concepts is that we introduce a different mathematical system, yet it yields the same results. Its introduction arises from the fact that the scalar and vector potentials from eqs. (2) and (3) are not unique. One can choose different varieties of ϕ and $\mathbf{A}$ and still produce same–valued $\mathbf{E}$ and $\mathbf{B}$ fields. The simplest idea behind a *transformation* is denoted by prime–notation. Applying this to eqs. (2) and (3), we transform both the $\mathbf{E}$ and $\mathbf{B}$ fields to the following:

$$\mathbf{E} = \nabla\phi' - \frac{\partial A'}{\partial t} \tag{6}$$

$$\mathbf{B} = \nabla \times \mathbf{A}' \tag{7}$$

What is left to do is figure out what that transformation is in order to preserve $\mathbf{E}$ and $\mathbf{B}$ fields of the same value. Let the transformation be defined as:

$$\mathbf{A}' = \mathbf{A} + \nabla\lambda \tag{8}$$

The reason behind this transformation is due to the rule regarding the curl of a gradient. It ends up being 0, thus providing us with a way to alter $\mathbf{A}$ while keeping $\mathbf{B}$ identical. Note we have the freedom to pick what λ is. To verify all this, we can substitute (8) into (3):

$$\begin{aligned}\mathbf{B} &= \nabla \times \mathbf{A}' \\ &= \nabla \times (\mathbf{A} + \nabla\lambda) \\ &= (\nabla \times \mathbf{A}) + (\nabla \times (\nabla\lambda)) \\ &= \nabla \times \mathbf{A} + 0 \\ &= \nabla \times \mathbf{A} \\ &= \mathbf{B}\end{aligned}$$

$\blacksquare$

Applying (8) to (2) yields:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial \mathbf{A}'}{\partial t} \\ &= -\nabla\phi - \frac{\partial}{\partial t}(\mathbf{A} + \nabla\lambda) \\ &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} - \frac{\partial}{\partial t}(\nabla\lambda) \\ &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} - \nabla\left(\frac{\partial\lambda}{\partial t}\right)\end{aligned}$$

We see that the 3rd term on the RHS of the equation is the reason as to why we aren't yielding identical results. Therefore, let our transformation of ϕ be:

$$\phi' = \phi - \frac{\partial\lambda}{\partial t} \tag{9}$$

Substitution of eqs. (8) and (9) into (2) yield:

$$\begin{aligned}\mathbf{E}' &= -\nabla\phi' - \frac{\partial \mathbf{A}'}{\partial t} \\ &= -\nabla\left(\phi - \frac{\partial\lambda}{\partial t}\right) - \frac{\partial}{\partial t}(\mathbf{A} + \nabla\lambda) \\ &= -\nabla\phi + \nabla\left(\frac{\partial\lambda}{\partial t}\right) - \frac{\partial \mathbf{A}}{\partial t} - \frac{\partial}{\partial t}(\nabla\lambda) \\ &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} + \nabla\left(\frac{\partial\lambda}{\partial t}\right) - \nabla\left(\frac{\partial\lambda}{\partial t}\right) \\ &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \\ &= \mathbf{E}\end{aligned}$$

$\blacksquare$

Finding a λ that satisfies eqs. (8) and (9) makes the scenario gauge transformative. Notably, when **E** and **B** are same–valued under these *gauge transformations*, it is known as *gauge invariance*.

4. Satisfying the Schrödinger Equation using Gauge Transformations

A proof of failure determines the need to define a transformed wavefunction Ψ'. Using the Time-Dependent Schrödinger Equation (TDSE) with minimal coupling:

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[\frac{1}{2m}(-i\hbar\nabla - q\mathbf{A}')^2 + q\varphi'\right]\Psi \tag{10}$$

and applying the gauge transformations from §1:

$$\mathbf{A}' = \mathbf{A} + \nabla\lambda$$
$$\varphi' = \varphi - \frac{\partial\lambda}{\partial t}$$

we substitute these directly into the Hamiltonian operator to test for form invariance under the assumption that Ψ remains unchanged.

First, we isolate the kinetic term operator inside the square brackets and expand it strictly by FOIL distribution from left to right, maintaining operator order:

$$\begin{aligned}(-i\hbar\nabla - q\mathbf{A}')^2 &= (-i\hbar\nabla - q\mathbf{A}') \cdot (-i\hbar\nabla - q\mathbf{A}') \\ &= (-\hbar^2\nabla^2) + (i\hbar q\nabla \cdot \mathbf{A}') + (i\hbar q\mathbf{A}' \cdot \nabla) + (q^2(\mathbf{A}')^2)\end{aligned}$$

Next, we substitute the definition $\mathbf{A}' = \mathbf{A} + \nabla\lambda$ directly into each of these four distributed operator pieces:

1. **First Term:** $-\hbar^2\nabla^2$ remains unchanged.
2. **Outer Term:** $i\hbar q\nabla \cdot (\mathbf{A} + \nabla\lambda) = i\hbar q(\nabla \cdot \mathbf{A}) + i\hbar q\nabla^2\lambda$
3. **Inner Term:** $i\hbar q(\mathbf{A} + \nabla\lambda) \cdot \nabla = i\hbar q(\mathbf{A} \cdot \nabla) + i\hbar q(\nabla\lambda \cdot \nabla)$
4. **Last Term:** $q^2(\mathbf{A} + \nabla\lambda)^2 = q^2\mathbf{A}^2 + 2q^2(\mathbf{A} \cdot \nabla\lambda) + q^2(\nabla\lambda)^2$

Reassembling these components with the scalar potential term $q\varphi' = q\varphi - q\frac{\partial\lambda}{\partial t}$, the complete expanded operator equation acting on Ψ yields:

$$\begin{aligned}i\hbar\frac{\partial\Psi}{\partial t} = \frac{1}{2m}\Big[&-\hbar^2\nabla^2 + i\hbar q(\nabla \cdot \mathbf{A}) + i\hbar q\nabla^2\lambda + i\hbar q(\mathbf{A} \cdot \nabla) \\ &+ i\hbar q(\nabla\lambda \cdot \nabla) + q^2\mathbf{A}^2 + 2q^2(\mathbf{A} \cdot \nabla\lambda) + q^2(\nabla\lambda)^2\Big]\Psi + q\varphi\Psi - q\frac{\partial\lambda}{\partial t}\Psi\end{aligned}$$

We now distribute Ψ across every term. We must evaluate the outer gradient term $i\hbar q\nabla \cdot (\mathbf{A}\Psi)$ using the vector derivative product rule $\nabla \cdot (\mathbf{V}f) = (\nabla \cdot \mathbf{V})f + \mathbf{V} \cdot \nabla f$:

$$i\hbar q\nabla \cdot (\mathbf{A}\Psi) = i\hbar q(\nabla \cdot \mathbf{A})\Psi + i\hbar q\mathbf{A} \cdot (\nabla\Psi)$$

Applying this same product rule logic to the operator term containing the gauge variable λ yields:

$$i\hbar q\nabla \cdot ((\nabla\lambda)\Psi) = i\hbar q(\nabla^2\lambda)\Psi + i\hbar q\nabla\lambda \cdot (\nabla\Psi)$$

Collecting all distributed components, we group the original unchanged TDSE terms together on the right-hand side, leaving the leftover gauge remnants isolated:

$$\begin{aligned}i\hbar\frac{\partial\Psi}{\partial t} = \left\{\frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2 + q\varphi\right\}\Psi + \frac{1}{2m}\Big[&i\hbar q(\nabla^2\lambda) + 2i\hbar q\nabla\lambda \cdot \nabla \\ &+ 2q^2(\mathbf{A} \cdot \nabla\lambda) + q^2(\nabla\lambda)^2\Big]\Psi - q\frac{\partial\lambda}{\partial t}\Psi\end{aligned}$$

Discussion of Failure

The resulting equation demonstrates that the TDSE is not satisfied "as is" under a gauge transformation. For form invariance to hold, the entire bracketed remnant must vanish. However, because $\lambda(\mathbf{r}, t)$ is an arbitrary space-time-dependent scalar function, these extra derivatives of λ and its cross-products with $\mathbf{A}$ are generally non-zero.

This mathematical mismatch explicitly determines that the physical state cannot be represented by the un-transformed wavefunction Ψ in the new gauge. To restore balance and cancel out this exact remainder, the wavefunction must undergo a local phase transformation of the form:

$$\Psi' = \Psi e^{i\frac{q}{\hbar}\lambda} \tag{11}$$

To restore form invariance and prove that TDSE = TDSE, we evaluate the transformed Time-Dependent Schrödinger Equation:

$$i\hbar\frac{\partial \Psi'}{\partial t} = \left[\frac{1}{2m}(-i\hbar\nabla - q\mathbf{A}')^2 + q\varphi'\right]\Psi' \tag{12}$$

by substituting the full set of gauge transformations directly into both sides simultaneously:

$$\begin{aligned}\mathbf{A}' &= \mathbf{A} + \nabla\lambda \\ \varphi' &= \varphi - \frac{\partial\lambda}{\partial t} \\ \Psi' &= \Psi e^{i\frac{q}{\hbar}\lambda}\end{aligned}$$

Expanding the left-hand side time derivative using the calculus product rule yields:

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\left(\Psi e^{i\frac{q}{\hbar}\lambda}\right) &= i\hbar\left(\frac{\partial\Psi}{\partial t}e^{i\frac{q}{\hbar}\lambda} + \Psi\cdot\left[i\frac{q}{\hbar}\frac{\partial\lambda}{\partial t}\right]e^{i\frac{q}{\hbar}\lambda}\right) \\ &= \left(i\hbar\frac{\partial\Psi}{\partial t} - q\frac{\partial\lambda}{\partial t}\Psi\right)e^{i\frac{q}{\hbar}\lambda}\end{aligned}$$

Next, we evaluate the action of the spatial kinetic operator on the transformed wavefunction. Expanding the first link of the derivative chain via the vector product rule gives:

$$\begin{aligned}(-i\hbar\nabla - q\mathbf{A}')\Psi' &= (-i\hbar\nabla - q\mathbf{A} - q\nabla\lambda)\left(\Psi e^{i\frac{q}{\hbar}\lambda}\right) \\ &= -i\hbar(\nabla\Psi)e^{i\frac{q}{\hbar}\lambda} - i\hbar\Psi\left(i\frac{q}{\hbar}\nabla\lambda\right)e^{i\frac{q}{\hbar}\lambda} - q\mathbf{A}\Psi e^{i\frac{q}{\hbar}\lambda} - q(\nabla\lambda)\Psi e^{i\frac{q}{\hbar}\lambda} \\ &= -i\hbar(\nabla\Psi)e^{i\frac{q}{\hbar}\lambda} + q(\nabla\lambda)\Psi e^{i\frac{q}{\hbar}\lambda} - q\mathbf{A}\Psi e^{i\frac{q}{\hbar}\lambda} - q(\nabla\lambda)\Psi e^{i\frac{q}{\hbar}\lambda} \\ &= [(-i\hbar\nabla - q\mathbf{A})\Psi]\, e^{i\frac{q}{\hbar}\lambda}\end{aligned}$$

Applying the second spatial operator to this intermediate result follows the identical algebraic pattern, pulling the exponential factor cleanly to the right of the squared operator:

$$\begin{aligned}(-i\hbar\nabla - q\mathbf{A}')^2\Psi' &= (-i\hbar\nabla - q\mathbf{A} - q\nabla\lambda)\cdot\left\{[(-i\hbar\nabla - q\mathbf{A})\Psi]\, e^{i\frac{q}{\hbar}\lambda}\right\} \\ &= \left[(-i\hbar\nabla - q\mathbf{A})^2\Psi\right]e^{i\frac{q}{\hbar}\lambda}\end{aligned}$$

Reassembling the entire equation by combining the expanded time derivative, the squared kinetic term, and the transformed scalar potential gives:

$$\left(i\hbar\frac{\partial\Psi}{\partial t} - q\frac{\partial\lambda}{\partial t}\Psi\right)e^{i\frac{q}{\hbar}\lambda} = \frac{1}{2m}\left[(-i\hbar\nabla - q\mathbf{A})^2\Psi\right]e^{i\frac{q}{\hbar}\lambda} + \left(q\varphi - q\frac{\partial\lambda}{\partial t}\right)\Psi e^{i\frac{q}{\hbar}\lambda}$$

Since $e^{i\frac{q}{\hbar}\lambda} \neq 0$, we divide through by the phase factor across all terms:

$$i\hbar\frac{\partial\Psi}{\partial t} - q\frac{\partial\lambda}{\partial t}\Psi = \frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2\Psi + q\varphi\Psi - q\frac{\partial\lambda}{\partial t}\Psi$$

Canceling the identical gauge remainder $-q\frac{\partial\lambda}{\partial t}\Psi$ from both sides restores the exact structure of the original equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[\frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2 + q\varphi\right]\Psi \tag{13}$$

The equation successfully reduces to its initial state, proving absolute form invariance (TDSE = TDSE). ■

5. Conclusion

This mathematical resolution carries profound physical consequences, serving as the foundational mechanism for the Aharonov-Bohm effect. It explicitly demonstrates that the vector potential $\mathbf{A}$ directly modulates the quantum behavior of a charged particle, even when the particle is strictly confined to a field-free region where the magnetic field $\mathbf{B}$ vanishes entirely.

This establishes a fundamental departure from classical electrodynamics. In classical mechanics, physical observables depend solely on localized force fields ($\mathbf{E}$ and $\mathbf{B}$), leaving potentials to be treated as mere mathematical conveniences. In the quantum regime, however, the gauge field itself acts non-locally upon the wavefunction. The local phase shift $e^{i\frac{q}{\hbar}\lambda}$ proves that the underlying potential possesses a physical reality that directly dictates quantum interference, rendering the concept of local fields insufficient to fully describe quantum dynamics.

References

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